EPFL Formal Verification Course Exam, 28 November 2024

IMPORTANT INFORMATION

Do not open the exam until we tell you to. The exam is three hours long. Place your CAMIPRO card on your desk. Put all electronic devices in a bag away from bench. Write using permanent, dark pen (no graphite nor heat-disappearing pen unless you were granted an exception). Write answers to different problems on disjoint sheets of paper that we supply. Write your name, SCIPER and question number on the top-right of each sheet you return. Do not write the solutions that you want us to grade on the sheets with exam questions; please take these printed exam sheets with you after the exam. You are only allowed two A4-sized two-sided cheat sheets (four A4 pages in total).

The maximal number of points on the exam is 40. We advise you to first solve questions that you find easier. If you are running out of time on a particular problem, try to convince us that you know the right strategy to solve it.

Reminder: If $t \subseteq B \times B$ is a binary relation and $C \subseteq B$, we define

$$t[C] = \{y \mid \exists x \in C.\ (x, y) \in t\}$$

The diagonal relation on $C \subseteq B$ is $\Delta_C = \{(x, x) \mid x \in C\}$. Given $t_1, t_2 \subseteq B \times B$, relation composition $\circ$ is:

$$t_1 \circ t_2 = \{(x, z) \mid \exists y.(x, y) \in t_1 \land (y, z) \in t_2\}$$

Reflexive transitive closure of t is

$$t^* = \bigcup_{i \geq 0} t^i$$

where $t^0 = \Delta_B$, $t^1 = t$, $t^{n+1} = t \circ t^n$. If $M = (S, I, r, A)$ is a transition system we define

$$\bar{r} = \{(s, s') \mid \exists a \in A.(s, a, s') \in r\}$$

and define the reachable states as image of I under the reflexive transitive closure of $\bar{r}$:

$$Reach(M) = \bar{r}^*[I]$$

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1 Lattices (10 points)

Let $(L, \sqsubseteq, \sqcup, \sqcap)$ be a lattice with the least element $\bot$. Suppose that L is such that every increasing chain of elements indexed by natural numbers, $x_0 \sqsubseteq x_1 \sqsubseteq \ldots$ has a least upper bound $\bigsqcup_{i \in \mathbb{N}} x_i$ in L. (You may not assume that the lattice is complete unless you can prove that first.) Let $f : L \to L$ be a function that is monotonic with respect to $\sqsubseteq$. Define the sequence $a_i \in L$, for $i \in \mathbb{N}$ inductively by:

$$\begin{aligned} a_0 &= \bot \\ a_{i+1} &= f(a_i) \end{aligned}$$

Define $g : L \to L$ by $g(x) = x \sqcup f(x)$ and the sequence $b_i \in L$, $i \in \mathbb{N}$ by:

$$\begin{aligned} b_0 &= \bot \\ b_{i+1} &= g(b_i) \end{aligned}$$

Let $\mathsf{Fix}_f, \mathsf{Fix}_g \subseteq L$ be the sets of fixed points of f and g, respectively. Let $\mathsf{Post}_f \subseteq L$ be the set of postfix points of f, that is, $\mathsf{Post}_f = \{x \mid f(x) \sqsubseteq x\}$.

Prove or give a counterexample for the following properties.

1. (1pt) g is monotonic with respect to $\sqsubseteq$.

2. (2pt) $\forall i \in \mathbb{N}.\ a_i = b_i$.

3. (1pt) $\forall x \in L.\ x \sqsubseteq g(x)$.

4. (1pt) $\mathsf{Fix}_g = \mathsf{Post}_f$.

5. (2pt) If f is ω-continuous then g is also ω-continuous.

6. (2pt) If f is ω-continuous then both f and g have the least fixpoint and these fixpoints are equal.

7. (1pt) For $L = \{\bot, \top\}$ with $\bot \sqsubseteq \top$ we have $\mathsf{Fix}_f = \mathsf{Fix}_g$.

2 Quantifier Elimination (5 points)

Use quantifier elimination to find and simplify a quantifier-free Presburger arithmetic formula equivalent to the following one:

$$\forall y.\ \big(\ 3|(y+1)\ \vee\ 3|(y+2)\ \vee\ y < x\ \vee\ 2x \leq y\big)$$

Show key steps. All variables range over integers. You may use any valid equivalence-preserving transformations on formulas interpreted over integers, but you need to state the equivalence rule you use explicitly. You may use parametric conjunctions and disjunctions in your intermediate steps ($\bigwedge_{i=\ldots}$, $\bigvee_{i=\ldots}$) but your final result should be written without such parametric operators, and should be as short as possible.

3 Resolution Proof (5 points)

Let R be a predicate of arity two in first-order logic. We say that a formula is closed it has no free variables.

1. (1pt) Write a closed FOL formula stating that R denotes a transitive relation: if x is in relation to y and y is in relation to z, then also x is in relation to z. Write a FOL clause corresponding to the formula above, denote it by A1.

2. (1pt) Write a closed FOL formula stating that R denotes an irreleflexive relation: no element is related to itself. Write a FOL clause corresponding to the formula above, denote it by A2.

3. (1pt) Write a closed FOL formula stating that it is never the case that, simulaneously, x is in relation to y and y is in relation to x. Denote this formula by G.

4. (2pt) Write a resolution proof that uses appropriate clauses and resolution rule steps to establish that the formula $A1 \wedge A2 \rightarrow G$ is valid. Each step should indicate which clauses it uses and which variable substitutions (if any) it applies to them.

4 Abstract Interpretation (5 points)

Consider the domain of integers intervals as a more concrete domain C, that is

$$C = \{\bot\} \cup \{[n, m] \mid n, m \in \mathbb{Z} \cup \{-\inf, +\inf\} \wedge n \leq m\}$$

and the domain of constant propagation as the abstract domain A, that is

$$A = \{\top, \bot\} \cup \{\{n\} \mid n \in \mathbb{Z}\}$$

Let $\alpha : C \to A$ and $\gamma : A \to C$ be given by

$$\alpha([n, n]) = \{n\}, \quad \alpha([n, m]) = \top \text{ for } n < m, \quad \alpha(\bot) = \bot$$

$$\gamma(\{n\}) = [n, n], \quad \gamma(\top) = [-inf, +inf], \quad \gamma(\bot) = \bot$$

Let $\leq$ denote the order on C where $\bot$ is the smallest element and otherwise $\leq$ is given by interval inclusion.

Let $\sqsubseteq$ denote the order on A where $\bot$ is the smallest element, $\top$ the largest, and all other elements are unrelated.

1. (1pt) Prove that α is monotonic.
2. (1pt) Prove that γ is monotonic.
3. (1pt) Prove that the defining condition of the Galois connection holds.
4. (1pt) Is function α injective? Prove or give a counterexample.
5. (1pt) Is function γ injective? Prove or give a counterexample.

5 Reachability (5 points)

Let $M = (S, I, r, A)$ be a transition system, $\bar{r} \subseteq S \times S$ defined as usual, and $G \subseteq S$ represent the set of good states. Let $D = 2^S \times 2^S$ (hence, D contains pairs of sets of states). Define $f : D \to D$ by

$$f((F, B)) = (F \cup sp(F, \bar{r}), B \cap wp(\bar{r}, B))$$

We will say that $(F, B) \in D$ is *safe* iff $F \subseteq B$. We will say that (F, B) is k-safe iff $f^k((F, B))$ is safe where we define, as usual,

$$f^0((F, B)) = (F, B), \qquad\qquad f^{n+1}((F, B)) = f(f^n((F, B)))$$

Your tasks:

1. (1pt) Define a partial order on D such that f is a monotonic function with respect to this order.

2. (2pt) Show that if G is an invariant, then (I, G) is k-safe for all k.
   Hint: Let x and y be respectively members of the first and second element of $f^n((F, B))$. What can we state about traces of length n whose final state is x? What can we say about y?

3. (2pt) Let n be the number of states in S. What is the smallest k as a function of n such that G is an invariant iff (I, G) is k-safe?

6 Tseitin's Transformation (4 points)

Consider a propositional formula F in negation normal form, using only the connectives $\wedge, \vee, \neg$. Let the free variables of F be $p_1, \ldots, p_n$. Let $T(F)$ denote the result of Tseitin's transformation that replaces a subformula that is a conjunction or disjunction of literals $l_1 * l_2$ (where $*$ denotes "$\vee$" or "$\wedge$") by a new propositional variable q_i and introduces clauses corresponding to the propositional equivalence $q_i = l_1 * l_2$. Suppose that the free variables of $T(F)$ are $p_1, \ldots, p_n, q_1, \ldots, q_m$.

- (1pt) Given the best upper bound (over all possible F) on the number of assignments of $\{0, 1\}$ to variables $p_1, \ldots, p_n$ that make F true.
- (1pt) Give the best upper bound (over all possible F) on the number of assignments of $\{0, 1\}$ to variables $p_1, \ldots, p_n, q_1, \ldots, q_m$ that make $T(F)$ true.
- (1pt) There is a variant T_1 of Tseitin's transformation that introduces clauses corresponding to an implication, instead of the propositional equivalence $q_i = (l_1 * l_2)$ and still generates an equisatisfiable formula. What is the direction of the implication that should be used for the introduced clauses, $q_i \to (l_1 * l_2)$ or $(l_1 * l_2) \to q_i$?
- (1pt) Give an upper bound on the number of satisfying assignments for the result of such optimized Tseitin's transformation assuming it generates a formula $T_1(F)$ with variables $p_1, \ldots, p_n, q_1, \ldots q_m$.

7 Predicate Abstraction (6 points)

Consider the following program over mathematical (unbounded) integers:

```
def myLittleProgram(var x: BigInt, var y: BigInt): Int =
  require(x >= 1 && y >= 1) ⓪
  x = x + y
  while ① (x >= y + 3) {
    ②
    if x % 2 == 0 then ③
      y = y + 1 ④
      x = x / 2 − 1 ⑤
    else ⑥
      x = x + 1 ⑦
      y = 0 ⑧
  }
  ⑨ y = x ∗ y
  ⑩ 1 / y
```

Note that / denotes integer division, rounded down. We identify program points using numbers ⓪,①,②,... Consider this set of 6 predicates:

$$\mathcal{P} = \{x \geq 1, x \geq 3, y \geq 0, y \geq 1, (y \geq 1 \lor x \geq 3), x\%2 = 0\}$$

1. (1pt) Let $\mathcal{P}' \subseteq \mathcal{P} = \{x \geq 1, x \geq 3, y \geq 0, y \geq 1\}$. Draw the Hasse diagram for the lattice $2^{\mathcal{P}'}$ of conjuncts over $\mathcal{P}'$ ordered by the predicate abstraction lattice order.

2. (1pt) Note that we did not include the predicate "false" in $\mathcal{P}$, but predicate abstraction should still work. Let the abstraction function from sets of states to sets of predicates to be α. Compute $\alpha(\emptyset)$ and $\gamma(\alpha(\emptyset))$.

3. (4pt) Using predicate abstraction as abstract interpretation, compute, for each indicated program point of myLittleProgram, the set of predicates from $\mathcal{P}$ that hold at that point. Only show the final conjunction of predicates for each point, not the entire iterative process.
   Does the annotation before the last line contain the predicate $y \geq 1$?